

D493

Scripting and Programming

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A script for a shipping company that processes 8,000 labels a day contains the line `code = 'AB12'` and then the line `code[0] = 'X'`. The developer wrote it on a new laptop. What happens when the second line runs?

- A. code becomes 'X'.
- B. A `TypeError`, because strings do not support item assignment.
- C. It depends on the new laptop's operating system.
- D. code becomes 'XB12'.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A warehouse script packs 17 items into boxes that hold 5 each. The warehouse has 3 loading docks. The script prints `17 // 5` and then `17 % 5`. What two values print?

- A. 3.4 and 2
- B. 3 and 3
- C. 2 and 3
- D. 3 and 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

In Python, `x // 4 == 6` and `x % 4 == 3`. What is x?

- A. 27
- B. 24
- C. 13
- D. 18

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A payroll script stores hours = [38, 40, 42, 45, 36] for five employees who all started in 2020. What does hours[1:4] return?

- A. [40, 42, 45]
- B. [40, 42, 45, 36]
- C. [38, 40, 42, 45]
- D. [36, 45, 42]

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Given items = ['a', 'b', 'c', 'd', 'e', 'f'], which slice returns ['c', 'd']?

- A. items[3:5]
- B. items[4:2]
- C. items[2:3]
- D. items[2:4]

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A backup script for a law office with 12 servers runs for i in range(2, 10, 3): print(i). Which values print?

- A. 3, 6, 9
- B. 2, 5, 8
- C. 10, 7, 4
- D. 2, 5, 8, 11

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which call produces exactly 10, 8, 6, 4?

- A. range(10, 4, -2)
- B. range(4, 10, 2)
- C. range(10, 3, -2)
- D. range(10, 2, 2)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A monitoring script sets `count = 0` at the top of the file. Inside a function it runs `count = count + 1` with no global statement. The script was written for a team of four admins. What happens when the function is called?

- A. The global count becomes 1.
- B. It depends on how many admins run the script.
- C. A new local count becomes 1 and the global stays 0.
- D. An `UnboundLocalError` is raised.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A file-import script for a clinic with 3 locations wraps `open()` in a try block, with except `FileNotFoundError` that prints “missing” and a finally block that prints “done.” The file exists and opens without error. What prints?

- A. “missing” and then “done”
- B. Only “missing,” because except blocks always run.
- C. Only “done”
- D. Nothing, since finally runs only after an error.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Which statement about a try / except / finally block is correct?

- A. An except clause naming one error catches every error type.
- B. The finally block runs whether or not an exception occurred.
- C. The else block runs when an exception occurs.
- D. The finally block runs only when an exception occurs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An inventory script at a hardware store with 20,000 products has `stock = {'bolt': 40, 'nut': 55}`. What does `stock.get('washer', 0)` return?

- A. 55
- B. None
- C. 0
- D. A `KeyError`.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Two Python projects on one analyst's workstation need different versions of the same library. The workstation has 32 GB of RAM. What is the standard fix?

- A.** Create a separate virtual environment for each project and install the needed version in each.
- B.** Add more RAM so both versions fit.
- C.** Always use the newest version, which works with all older code.
- D.** Install one version globally and let the other project use it too.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A script for a shipping company ~~that processes 8,000 labels a day~~ contains the line code = 'AB12' and then the line code[0] = 'X'. ~~The developer wrote it on a new laptop.~~ What happens when the second line runs?

Target word: result of item assignment on a string **Decides:** Strings are immutable.

A	Dropped element	Treats the line as replacing the whole value.
B	Correct	Strings cannot be changed in place; build a new string instead.
C	Camouflage	Immutability is part of the language on every platform.
D	Confusion set	That is how a list would behave, not a string.

2. Answer D Concept 2

A warehouse script packs 17 items into boxes that hold 5 each. ~~The warehouse has 3 loading docks.~~ The script prints 17 // 5 and then 17 % 5. What two values print?

Target word: floor division and remainder **Decides:** $17 = 3 \times 5 + 2$.

A	Confusion set	3.4 is 17 / 5; // returns the whole quotient.
B	Wrong stem number	Uses the dock count as the remainder.
C	Inverted operation	Swaps the quotient and the remainder.
D	Correct	Three full boxes, two items left over.

3. Answer A Concept 2 · transfer

In Python, $x // 4 == 6$ and $x \% 4 == 3$. What is x?

Target word: solve for the dividend **Decides:** $x = 4 \times 6 + 3$.

A	Correct	$27 // 4 = 6$ and $27 \% 4 = 3$.
B	Dropped element	Forgets to add the remainder.
C	Wrong stem number	Just adds the numbers in the stem.
D	Inverted operation	$4 \times 3 + 6$ swaps the roles of quotient and remainder.

4. Answer A Concept 3

A payroll script stores hours = [38, 40, 42, 45, 36] for five employees ~~who all started in 2020~~. What does hours[1:4] return?

Target word: slice result **Decides:** Index 1 through 3; stop index 4 is excluded.

A	Correct	Indexes 1, 2, and 3.
B	Dropped element	Includes index 4, dropping the stop-is-excluded rule.
C	Confusion set	Counts from 1 and includes the stop.
D	Inverted operation	Reads from the end of the list.

5. Answer D Concept 3 · transfer

Given items = ['a', 'b', 'c', 'd', 'e', 'f'], which slice returns ['c', 'd']?

Target word: choose the slice — reverse **Decides:** 'c' is index 2; stop must be 4.

A	Confusion set	Counts from 1; returns ['d', 'e'].
B	Inverted operation	Start after stop returns an empty list.
C	Dropped element	Returns only ['c']; stop is excluded.
D	Correct	Indexes 2 and 3.

6. Answer B Concept 4

A backup script ~~for a law office with 12 servers~~ runs for i in range(2, 10, 3): print(i). Which values print?

Target word: range output **Decides:** Start 2, step 3, stop before 10.

A	Confusion set	Treats the step as the start.
B	Correct	The next value, 11, is past the stop.
C	Inverted operation	Counts down from the stop.
D	Dropped element	Ignores the stop value.

7. Answer C Concept 4 · transfer

Which call produces exactly 10, 8, 6, 4?

Target word: choose the range — reverse **Decides:** Negative step; stop must be below 4 and above 2.

A	Dropped element	Stops before 4, giving 10, 8, 6.
B	Inverted operation	Counts up: 4, 6, 8.
C	Correct	10, 8, 6, 4; the next value, 2, is past the stop of 3.
D	Adjacent category	A positive step can never go from 10 down to 2, so it is empty.

8. Answer D Concept 5

A monitoring script sets count = 0 at the top of the file. Inside a function it runs count = count + 1 with no global statement. ~~The script was written for a team of four admins.~~ What happens when the function is called?

Target word: scope outcome **Decides:** Assignment makes count local, and it is read before it is set.

A	Dropped element	Drops the need for a global declaration.
B	Camouflage	Scope rules do not depend on users.
C	Confusion set	The local is read before assignment, so it fails.
D	Correct	count is local to the function and has no value yet when read.

9. Answer C Concept 6

A file-import script ~~for a clinic with 3 locations~~ wraps open() in a try block, with except FileNotFoundError that prints “missing” and a finally block that prints “done.” The file exists and opens without error. What prints?

Target word: output when no exception occurs **Decides:** No exception, and finally always runs.

A	Confusion set	except runs only when the matching error occurs.
B	Absolute language	except is conditional.
C	Correct	except is skipped; finally runs.
D	Inverted operation	finally runs every time.

10. Answer B Concept 6 · transfer

Which statement about a try / except / finally block is correct?

Target word: rule statement **Decides:** finally always runs.

A	Absolute language	A named except catches only that type and its subclasses.
B	Correct	That is its purpose, e.g., cleanup.
C	Confusion set	else runs when no exception occurs.
D	Inverted operation	It runs in both cases.

11. Answer C Concept 7

An inventory script at a hardware store ~~with 20,000 products~~ has stock = {'bolt': 40, 'nut': 55}. What does stock.get('washer', 0) return?

Target word: get with a default **Decides:** The key is missing and a default of 0 is given.

A	Wrong stem number	The value of another key.
B	Dropped element	None is returned only when no default is supplied.
C	Correct	.get returns the default for a missing key.
D	Confusion set	That is stock['washer'], not .get.

12. Answer A Concept 8

Two Python projects on one analyst's workstation need different versions of the same library. ~~The workstation has 32 GB of RAM.~~ What is the standard fix?

Target word: dependency isolation **Decides:** Conflicting versions across projects.

A	Correct	Virtual environments isolate package versions per project.
B	Camouflage	Memory has nothing to do with version conflicts.
C	Absolute language	New versions can break older code.
D	Dropped element	Drops one project's requirement.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	1	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mutable vs. immutable types	Lists and dictionaries can be changed in place; strings, tuples, and numbers cannot.	Assigning to one character of a string as if it were a list.	Item assignment on a string or tuple.
2	Floor division and modulo	$a // b$ gives the whole-number quotient (floored); $a \% b$ gives the remainder; a / b always gives a float.	Mixing up $//$ and $/$, or quotient and remainder.	Boxes, pages, or groups with leftovers.
3	Indexing and slicing	Indexes start at 0; a slice <code>[start:stop]</code> includes start and excludes stop.	Counting from 1 or including the stop index.	A slice like <code>[1:4]</code> .
4	range() boundaries	<code>range(start, stop, step)</code> starts at start and stops before reaching stop.	Including the stop value, or reading the step as the start.	A for loop over range with three arguments.
5	Variable scope	Assigning to a name inside a function makes it local for the whole function unless declared global (or nonlocal).	Assuming a function can update a module-level variable just by assigning to it.	A top-level counter changed inside a function.
6	try / except / finally	<code>except</code> runs only when a matching exception occurs; <code>finally</code> runs every time.	Thinking <code>finally</code> runs only on errors, or that <code>except</code> always runs.	A try block where no error happens.
7	Dictionary lookups	<code>d[key]</code> raises <code>KeyError</code> for a missing key; <code>d.get(key, default)</code> returns the default.	Expecting <code>.get</code> to raise, or to return <code>None</code> when a default is supplied.	A lookup of a key that is not in the dictionary.
8	Virtual environments	Each project gets its own virtual environment so its packages and versions do not collide with other projects.	Installing everything globally and hoping versions agree.	Two projects needing different versions of one library.

Course scope (public description): Applications - Scripting and Programming - Applications for undergraduates explores the various aspects of the Python programming language by examining its syntax, the development environment, and tools and techniques to solve some real-world problems. Introduction to Programming in Python is a prerequisite for this course.